Customer Shopping Behavior Analysis

Turning retail transaction data into marketing and product decisions

1. Executive Summary

This project analyzes 3,900 retail transactions to uncover spending patterns, customer segments and product preferences. Python cleaned the data, MySQL answered the business questions, and Power BI presented the KPIs. Three findings stand out: only 27% of customers subscribe, Clothing leads revenue while Outerwear trails, and Young Adult and Middle-Aged customers are the most active buyers.

2. Business Problem

A large retailer wants to grow sales and loyalty, but purchasing patterns are shifting across demographics and sales channels. **Core question:** How can the company use shopping data to spot trends, engage customers and sharpen its marketing and product strategy?

3. Methodology

Data preparation (Python)

- Filled 37 missing Review Ratings with the median rating of each product category.
- Grouped customers into four age segments: young adult, adult, middle-aged and senior.
- Converted purchase frequency into days between purchases (Weekly = 7, Annually = 365).
- Dropped redundant columns such as promo_code_used and standardized column names for MySQL.

Analysis (SQL) and dashboard (Power BI)

- MySQL queries covered revenue, category performance, discounts and shipping.
- A Power BI dashboard tracks the headline KPIs for stakeholders.

4. Key Findings

- **KPIs:** 3.9K transactions, $59.76 average purchase, 3.75 average review rating, and a subscription rate of 27% (yes) vs 73% (no).
- **Revenue by gender:** Male customers generated $157,890 and female customers $75,191, so men account for about 68% of the roughly $233K total.
- **Categories:** Clothing is the main revenue driver; Outerwear contributes the least.
- **Discounts:** Hats (50%) and Sneakers (49.66%) have the highest share of discounted purchases.

- **Shipping:** Express customers spend slightly more ($60.48) than Standard customers ($58.46).

5. Recommendations

1. **Grow subscriptions:** With 73% of customers not subscribed, use targeted loyalty offers to build recurring revenue.
2. **Target key segments:** Prioritize Young Adult and Middle-Aged customers, the most active buyers.
3. **Lift Outerwear:** Test specialized promotions or targeted discounts for the weakest category.
4. **Improve quality:** Investigate low-rated products to raise satisfaction.
5. **Reward repeat buyers:** Launch a loyalty program for customers with more than 5 previous purchases.

6. Conclusion

This multi-tool analysis gives the company a practical roadmap for data-driven decisions. By converting non-subscribers and focusing on its highest-revenue customers, it can strengthen engagement and growth. Before acting, validate the gender revenue gap against customer counts and test the discount findings.

Python scripts, SQL queries and Power BI files are available in the project's GitHub repository.